JMORF/MORS — Morpho-Syntax

Grammar and Processing

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Lecture 12
Location: KB-1.11

Overview

- Grammar and psychology: competence, performance and acquisition
- Production
- Comprehension
- Lexical knowledge in processing
- Universals

Grammar and Psychology

What Does Grammar Have to Do with Psychology?

Three ways it could be relevant:

- It provides insight into how children acquire language.
- It provides insight into how speakers produce utterances.
- It provides insight into how listeners understand utterances.

Our Model: Key Characteristics

Surface-oriented

Constraint-based

Lexicalist

Chomsky's Position

- Grammar represents knowledge of language (**competence**).
- This is distinct from use of language (**performance**).
- We can draw a strong conclusion about language acquisition, namely, most grammatical knowledge is innate and task-specific.
- Serious study of language use (production and comprehension) depends on having a well-developed theory of competence.

Brief Remarks on Language Acquisition

- Chomsky's nativism is very controversial.
  - It is based on the **poverty of the stimulus** argument, and a model of learning as hypothesis testing.
  - The environment may be more informative than he assumes.
  - There may be more powerful learning methods than he assumes.

- There has not been much work on language acquisition using constraint-based lexicalist theories like ours; but
  - Explicit formulation is a prerequisite for testing learning models.
  - Our feature structures could model richer context information.

- We're neutral with respect to this controversy.

Production

Production and Grammar

- Evidence for left-to-right effects
- Evidence for top-down planning

Disfluencies Are Sensitive to Structure

Repeat rate of *the* varies with position and complexity of the NP it introduces:

- More common in front of complex NPs
- More common with prominent NPs
  Topic > Subject > Direct Object > Preposition Object

(1) *The the book I told the student about on the train*
(2) > *The book I told the the student about on the train*
(3) > *The book I told the student about on the the train*

Production Errors Are Sensitive to Syntactic Structure

- Agreement errors are more common with PP complements than sentential complements: errors like (5) are significantly more common than errors like (4).

  (4) **The claim that the wolves had raised the babies were rejected.*

  (5) > **The claim about the newborn babies were rejected.*

- Why?
  - Speculation: Clauses are their own agreement domains, so people don't mistake an NP in a lower clause as a trigger for agreement.
  - Original work: Kay Bock (1980s).

Some High-Level Sentence Planning Is Necessary, Too

(6) *Ich habe dem Mann, den ich gesehen habe, geholfen.*
I have the-dat man who-acc I seen have helped
"I helped the man I saw."

(7) *Ich habe den Mann, dem ich geholfen habe, gesehen.*
I have the-acc man who-dat I helped have seen
"I saw the man I helped."

- The choice between *dem* and *den* depends on the choice of verbs several words later.

Interaction of Top-Down and Left-to-Right Information

- Grammar plays a role in production.
- Partial grammatical information should be accessible by the production mechanism as needed.
- This argues against grammatical theories that involve sequential derivations with fixed ordering.
- Our theory of grammar has the requisite flexibility.

Comprehension

Comprehension

- Early work tried to use transformational grammar in modeling comprehension.
- **The Derivational Theory of Complexity**: The psychological complexity of a sentence increases with the number of transformations involved in its derivation.
- Initial results seemed promising, but later work falsified the DTC.

Some Relevant Quotes

- "The results show a remarkable correlation of amount of memory and number of transformations"

  — Chomsky (1968)

- "[I]nvestigations of DTC ... have generally proved equivocal. This argues against the occurrence of grammatical derivations in the computations involved in sentence recognition"

  — Fodor, Bever, & Garrett (1974)

- "Experimental investigations of the psychological reality of linguistic structural descriptions have ...proved quite successful."

  — Fodor, Bever, & Garrett (1974)

- In particular, they concluded that **deep structures** and **surface structures** were psychologically real, but the transformations relating them weren't.

Evidence for the Psychological Reality of Deep Structures

- The proposed Deep Structure for (9) had three occurrences of ***detective***, while the proposed DS for (8) had only two:

  (8) *The governor asked the detective to prevent drinking.*

  (9) *The governor asked the detective to cease drinking.*

- In a recall experiment, ***detective*** was significantly more effective in prompting people to remember (9) than (8).

Typical Problem Cases for the DTC

(10) *Pat swam faster than Chris swam.*

(11) *Pat swam faster than Chris did.*

(12) *Pat swam faster than Chris.*

- The DTC predicts that (10) should be less complex than (11) or (12), because (11) and (12) involve an extra deletion transformation.

- In fact, subjects responded more slowly to (10) than to either (11) or (12).

What Should a Psychologically Real Theory of Grammar Be Like?

- The **deep structure** distinctions that are not evident on the surface should be represented.
- The transformational operations relating deep and surface structures should not be part of the theory.
- Our information-rich trees include all of the essential information in the traditional deep structures, but without the transformations.

Jerry Fodor Claims the Human Mind Is modular

> "A module is ... an informationally encapsulated computational system — an inference-making mechanism whose access to background information is constrained by general features of cognitive architecture."
>
> — Fodor, 1985

- A central issue in psycholinguistics over the past 20 years has been whether language is processed in a modular fashion.

Tanenhaus's Eye-Tracking Experiments

- Participants wear a device on their heads that makes a videotape showing exactly what they're looking at.
- They listen to spoken instructions and carry out various tasks.
- The eye-tracking provides evidence of the cognitive activity of participants that can be correlated with the linguistic input.

Non-Linguistic Visual Information Affects Lexical Access

- Participants' gaze settled on a referent before the word was completed, unless the initial syllable of the word was consistent with more than one object.

- For example, participants' gaze rested on the pencil after hearing *Pick up the pencil* more slowly when both a **p**encil and a **p**enny were present.

Non-Linguistic Visual Information Affects Syntactic Processing

- Eye movements showed that people hearing (13) often temporarily misinterpreted *on the towel* as the destination.

  (13) *Put the apple on the towel in the box.*

- When *on the towel* helped them choose between two apples, such misparses were significantly less frequent than when there was only one apple.

General Conclusion of Eye-Tracking Studies

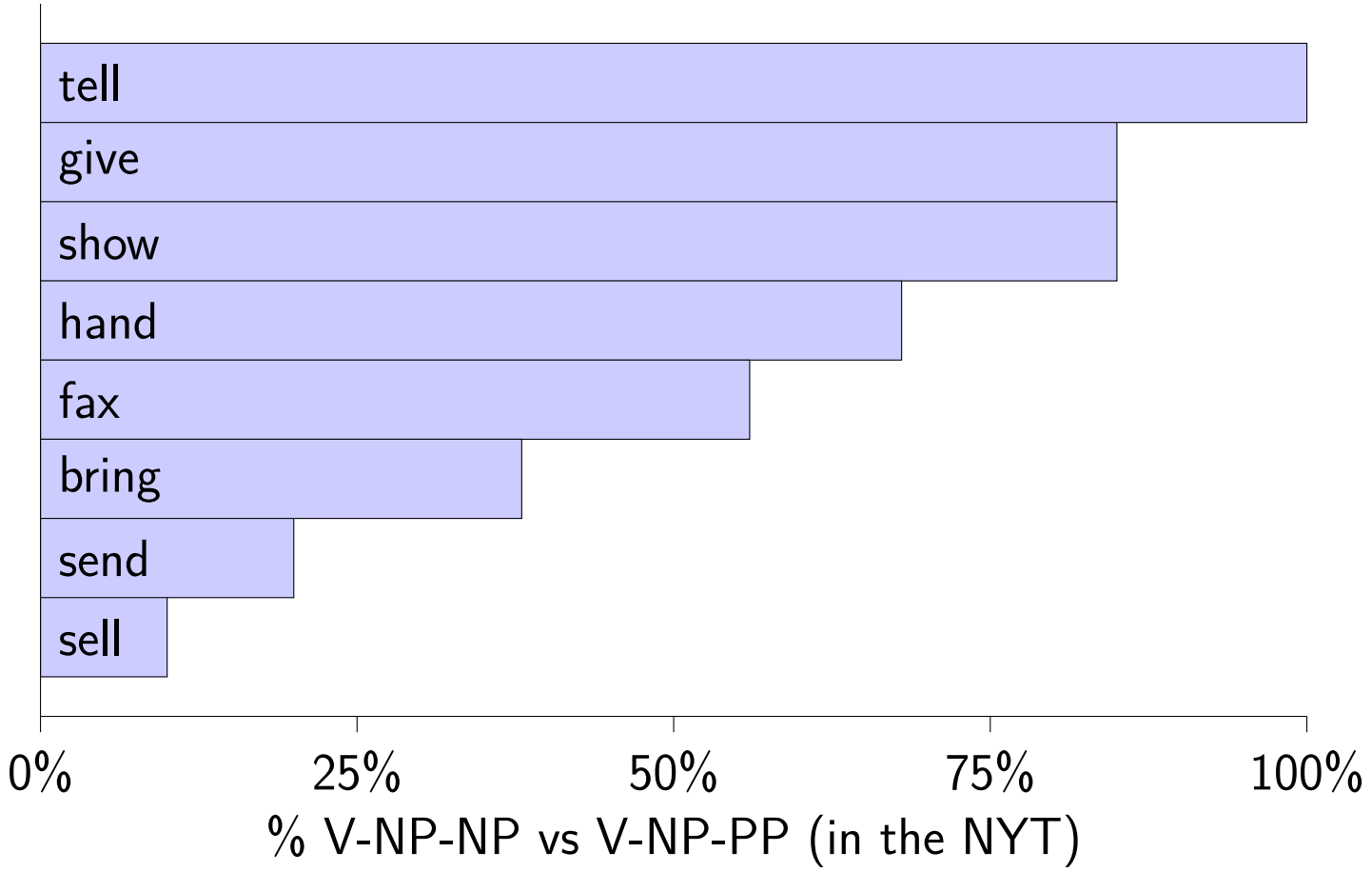
- People use whatever information is available as soon as it is useful in interpreting utterances.
- This argues against Fodorian modularity.
- It argues for a model of language in which information is represented in a uniform, order-independent fashion.

Lexical Knowledge in Processing

Speakers Know a Great Deal about Individual Words

- Individual lexical items have many idiosyncrasies in where they can occur, and in where they tend to occur.
- For example, the verb ***behoove*** occurs only with the subject ***it*** (and only in certain verb forms), and the verb ***beware*** has only the base form.
- We also know that the transitive use of ***walk*** is much rarer than the intransitive.

Different Verbs Favor Different COMPS Lists



Lexical Biases Influence Processing

- Wasow et al. ran a production experiment to test whether ambiguity avoidance would influence speakers' choice between (14) and (15):

  (14) *They gave Grant's letters to Lincoln to a museum.* NP-PP

  (15) *They gave a museum Grant's letters to Lincoln.* NP-NP

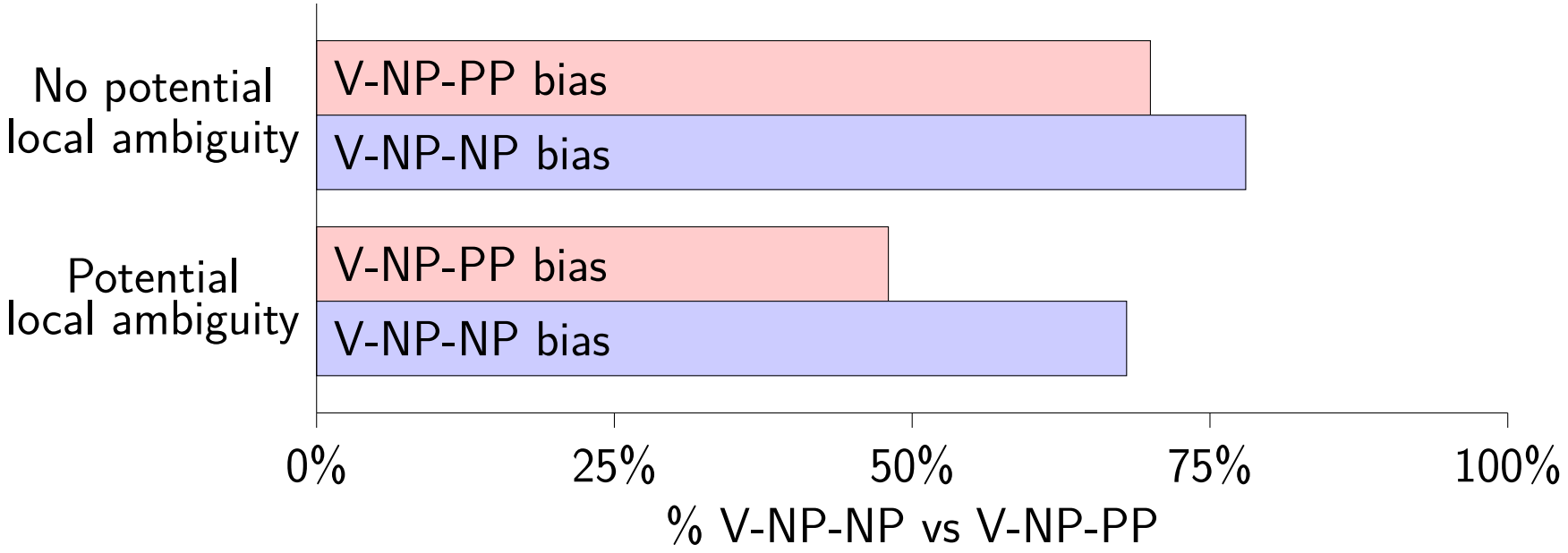
- Ambiguity avoidance predicts that you should prefer (15).

- Lexical bias of the verbs turned out to be a significant predictor of which form speakers used (and ambiguity avoidance turned out not to be).

Experimental Method

- Speaker and Listener sit next to each other. Speaker can see a screen.
- Speaker silently reads a sentence shown on the screen
  A museum in Philadelphia received Grant's letters to Lincoln from the foundation.
- The sentence disappears from the screen.
- Listener asks a question:
  What did the foundation do?
- The speaker answers the listener's question.
  The foundation gave ... the museum, um, Grant's letters to Lincoln.
- The listener records which kind of response on a list (from two choices).

Experimental Results on Local Ambiguity



- Arnold, Wasow, Asudeh & Alrenga (2004: *Journal of Memory & Language*) re-ran the experiment with slightly better methodology and found an even stronger reverse ambiguity effect.

A Psychologically Real Grammar Should Be Lexicalist

- Early generative grammars downplayed the lexicon.
- Now, however, the importance of the lexicon is widely recognized.
- This aspect of grammar has been developed in greater detail in HPSG than in any other theory.
- It would be easy to add frequency information to the lexicon, though there is debate over the wisdom of doing so.
- Frequency is currently recorded as part of the **parse ranking model**s that select the most plausible out of all possible interpretations.

Conclusion

- Grammatical theory should inform and be informed by psycholinguistic experimentation.
- This has happened less than it should have.
- Existing psycholinguistic evidence favors a constraint-based, lexicalist approach (like HPSG, LFG, construction grammar).

Universals

Universals?

- **Principles and Parameters (P&P)**: attempts to relate multiple typological properties to single parameters (top-down).

- **Grammar Matrix**: attempts to describe many languages in a consistent framework and then takes stock of common constraints (bottom-up).

  – Until we know more detail about more languages, we cannot test theories properly
  – So describing languages in precise detail is very important
  – HPSG is fully in the tradition of language documentation
  – The AGGREGATION Project[1] attempts to automate the construction of grammar fragments from language descriptions, building on interlinear glossed text (IGT) and using the Grammar Matrix

[1] Automatic Generation of Grammars for Endangered Languages from Glosses and Typological Information

Universals?

- Case Constraint
- SHAC
- Binding theory
- Head-Complement, Head-Specifier and Head-Modifier Rules
- Head Feature Principle
- Valence Principle
- Semantic Compositionality Principle
- ...

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Acknowledgments and References

- Course design and slides borrow heavily from Emily Bender's course: *Linguistics 566: Introduction to Syntax for Computational Linguistics* http://courses.washington.edu/ling566

- This lecture complements Sag et al. (2003, Chapter 9)

References

Ivan A. Sag, Tom Wasow, and Emily Bender. 2003. *Syntactic Theory: A Formal Introduction*. CSLI Publications, Stanford, second edition.